

1 A light bulb moment

Those of you living in metros probably haven't experienced India's dearth of electric power, and those of you from smaller towns are nodding your heads up and down vigorously in agreement – so hard, in fact, that chiropractors will laugh all the way to the bank this month.

For decades our biggest problem has been power, and the fact that India just does not generate enough to supply to its billion-plus population. Any technology you want to use needs power to run it.

Solution: **CAPP**

A cheap power source that can be used off the grid, or even to supply back to the grid... Sounds like sci-fi? All technology is sci-fi a decade or century before it's invented. We're going to call our tech CAPP (cheap and portable power).

How it could work

Now if there's one area where there's an embarrassment of riches when it comes to theory, that's power. From anti-matter generators to cold fusion and even everyday solutions like solar or hydroelectric energy, there are enough ways mankind knows how to create power. Drawbacks include too much power input and too little power output, or basically low efficiency generation techniques. Then there's the whole field of research that aims at reducing power consumption to begin with. Either way, there are tonnes of solutions to the problem, and what it really needs is all of us to start following it. Let's look at a few solutions that we feel are promising.

The ordinary

Conserve: The most obvious way to solve our power problems is to reduce the amount we use, and we've told you how to do exactly that in previous articles.

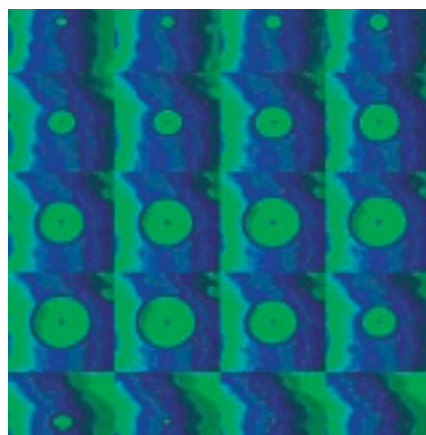
Sun, wind, water: Living in India has its advantages. The extreme summers can be put to good use, providing we find even cheaper ways of manufacturing photovoltaic cells (solar cells). A little less reliable than solar energy, wind power is still a viable source. Some innovative ideas suggest that we could have gliders that

ride the winds and collect energy as they do so, and then land to deliver it at base stations. The same applies to hydroelectric energy, which is again seasonal, but very usable in India's monsoons, which is also when solar power is least reliable.

The extraordinary

Sonoluminescence: A strange word indeed, but with perhaps the most promising yet. If only we can figure out how to get it to work. Basically, sonoluminescence is using sound (sono) to heat up a bubble or air in a liquid and produce light (luminescence). It might not sound like much, but the methods being used could theoretically solve the cold fusion problem. Going into detail about this phenomenon is out of the scope of this article, but all of you physics majors out there might want to look into it. In a nutshell, if you pass high-frequency sound waves (between 20 and 40 KHz) through a liquid that has an air bubble, the sound waves cause the bubble to contract suddenly, and briefly produce temperatures hotter than the surface of the sun at the centre of the bubble. That's, hot enough to start a "fusion" reaction, and is contained in a relatively "cold" liquid surrounding. If you're in the field, do the research, and remember us when you bag that Nobel prize.

Piezoelectricity: The one thing we do not have a shortage of as a country is people. Now consider our large cities, where millions reside, sometimes in spaces where perhaps only a few thousand should. Japan



A sequence of false colour images of growth and collapse of a single bubble in sulphuric acid irradiated with ultrasound



Prof. Yuji Ohya from Kyushu University, Japan, suggests we use floating windmills

is no different, but instead of sitting around complaining about the crowds, one Japanese scientist, Kohei Hayamizu, decided to put all those crowds to good use, and started using piezoelectric crystals to harness all our wasted energy. To illustrate, think of one of Mumbai's busy stations for instance. With literally lakhs of people walking up and down stairs and platforms every day, imagine if we could get a little energy from each one of them? Kohei Hayamizu's idea is simple – what if stairs, pavements, roads and anything else that objects or people move over, could generate electricity? You climb a staircase, and the stairs move ever so slightly, absorbing some of your energy. Of course it would mean that all of us would find such stairs just ever so slightly tougher to climb, thanks to the law of conservation of energy, but chances are we'd never notice it. Of course the innovation doesn't stop here. You could use piezoelectric cells to line your shirts, shoes, mobile phone keypad, and absolutely anything that moves. Cell phone service providers would kill for this today – imagine if everyone with low phone batteries started SMSing friends like crazy to build up some charge? On both, the battery and the phone bill, that is...

What the above ideas symbolize is a way to make electricity locally, for local needs. For India, and the world, any one of these could be our winning formula. Of course there are a hundred other ideas that could do the same here, such as smarter grids that are more efficient, and thus lose less energy, or more efficient circuits, and even super-conductors. The possibilities are endless, and all we need is one of them to work.

Must-clicks

<http://bit.ly/bMa7p9> | <http://bit.ly/dxYyMH>
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